

An LSTM-based approach to detect transition to lean blowout in swirl-stabilized dump combustion systems

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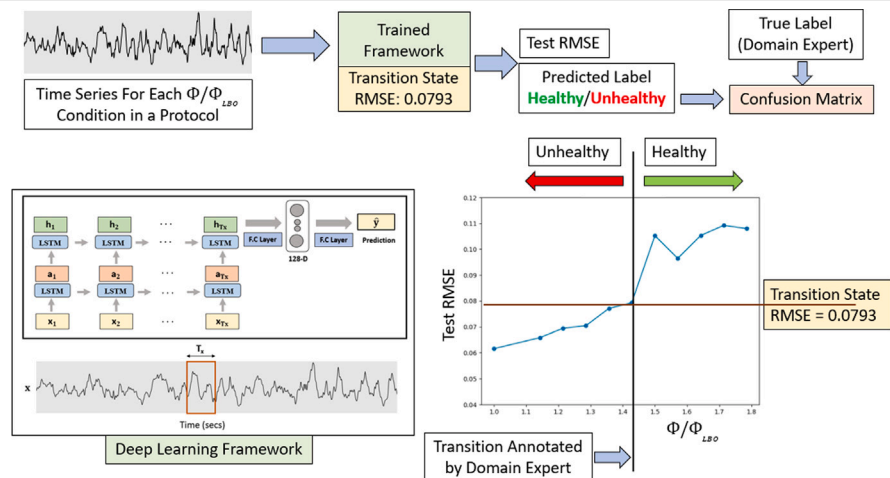
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HIGHLIGHTS

- Proposing a deep learning approach to detect lean blowout in combustion systems.
- Comparing the performance of the proposed framework against three baseline methods.
- More accurate, computationally faster, and captures the transition to a lean blowout regime better than baseline methods.
- Shows the potential to be implemented as an online detector to prevent the adverse effects of LBO.

GRAPHICAL ABSTRACT



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ABSTRACT

Lean combustion is environment friendly with low NO_x emissions providing better fuel efficiency in a combustion system. However, approaching towards lean combustion can make engines more susceptible to an undesirable phenomenon called lean blowout (LBO) that can cause flame extinction leading to sudden loss of power. During the design stage, it is quite challenging for the scientists to accurately determine the optimal operating limits to avoid sudden LBO occurrences. Therefore, it is crucial to develop accurate and computationally tractable frameworks for online LBO prediction in low NO_x emission engines. To the best of our knowledge, for the first time, we propose a deep learning approach to detect the transition to LBO in combustion systems. In this work, we utilize a laboratory-scale swirl-stabilized combustor to collect acoustic data for different protocols. For each protocol, starting far from LBO, we gradually move towards the LBO regime, capturing a quasi-static time series dataset at different conditions. Using one of the protocols in our

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dataset as the reference protocol, we find a transition state metric for our trained deep learning model to detect the imminent LBO in other test protocols. We find that our proposed approach is more precise and computationally faster than other baseline models to detect the transition to LBO. Therefore, we endorse this technique for monitoring the operation of lean combustion engines in real time.

1. Introduction

A primary disadvantage of using rich combustion systems in industrial and rocket propulsion applications is NO_x emissions, influenced by local high flame temperatures [1]. Such rich combustion systems include non-premixed or diffusion combustion and fuel-rich premixed combustion for a certain range of equivalence ratio above 1.0 (i.e., stoichiometric fuel-air condition). The necessity for having low NO_x combustion in the energy sectors, such as furnace or boilers, power plants and land-based engines, motivates scientists to move towards lean combustion [2–5]. In addition to reducing NO_x emissions, lean combustion improves fuel economy. Therefore, such a strategy aims at developing a cleaner and better environment.

However, during such ultra-lean operation regimes, the fuel-air ratio becomes very low, making the flame unstable, and the flame can even get extinguished by blowing out of the combustion chamber. Such an unwanted event in the operation of combustion systems is referred to as lean blowout (LBO) [6–9]. Generally, the probability of occurrence of LBO increases when the unburnt mixture velocity exceeds the reaction speed [4].

LBO may occur during scenarios like low fuel flow [8,10], highly turbulent combustion [11], and large variations in fuel flow [6]. The concept of LBO is illustrated in Fig. 1. Low fuel flow scenarios may occur when changes in the throttle settings of an aero-engine is necessary for deceleration. However, sometimes, sudden and inaccurate variation in throttle may rapidly reduce the fuel flow rate, leading to extremely low equivalence ratio as the rate of change of air flow is slower due to inertia of the compressor [12]. Extreme low fuel conditions can also arise in land-based engines during the adoption of an ultra-lean premixed combustion to avoid NO_x formation. On the other hand, a continuous increase in air flow rate while maintaining a constant power input can produce extremely turbulent combustion and, thereafter, LBO [13,14]. LBO can also be caused by large variations in the fuel flow, which can occur when there are sudden changes in power requirement [15].

As the lean combustors are used in a broad spectrum of industrial applications such as ground-based engines [16], aero-engines [17,18] and other gas turbine engines [19], the possibility of LBO can have significant impact on many such systems. For the ground-based engines in the power sector, LBO can cause unexpected stoppage of the engine, collapsing the entire power network [15]. Such disruptions of engine activity affect productivity and can cause significantly higher maintenance costs. On the other hand, for aviation, the risk of LBO is even higher when there is a low fuel flow scenario [12], and it can be challenging to recover from such a sudden flame loss during the running condition.

In the absence of any reliable LBO detection system, the standard practice in industry is to operate combustion engines with a high safety level. This implies operating engines at air-fuel ratios significantly greater than the LBO limit, resulting in increased NO_x emissions. For the aero-engines, the absence of adequate LBO detection and control facility decreases the ability to manoeuvre engines — a serious problem particularly for military aircrafts. Thus, the study of LBO has become important for the science community, especially when engineers want to deal with “clean energy” technology (e.g., lean combustion). As it is difficult to determine the LBO regimes during the design phase, developing an online prediction tool for LBO can be an important contribution to different application domains.

Previous studies in the context of LBO characterization and detection are mostly concentrated on the premixed type of combustion

(i.e., fixed degree of air-fuel mixing) [6,20–23]. From stable combustion to the blowout event, researchers distinguished different states for the premixed combustion, such as combustion noise, intermittency, thermoacoustic instability, and lean blowout [11]. Other studies detected a bifurcation in the behavior of combustion at which flame lifts off from its stability point [8,24] and found the extinction-reignition cycles [6,24] prior to the blowout event. Based on such behavioral changes of the premixed combustion, researchers have proposed different statistical [6,20,25] and nonlinear dynamics-based data-driven techniques [11,13,26]. However, most earlier works do not provide the details of computational tractability, a significant factor to understand the feasibility of implementing a technique for real-time performance monitoring of engines.

In aero-engines and modern land-based engines, instead of homogeneous fuel-air mixing, either fuel is injected close to the reaction chamber, or multiple fuel-air injections are used. Consequently, combustion produced from such a fuel-air configuration is no longer premixed and becomes partially premixed [27]. Therefore, an early prediction of the onset of LBO is also crucial for partially premixed combustion processes, where the degree of fuel-air mixing varies [24]. There has been a recent focus on LBO prediction in partially premixed combustion [8,9,24]. While statistical measures like mean, probability density function, and root mean square of combustion oscillations, are useful for partially premixed combustion, precursors based on extinction-reignition cycles are not beneficial due to the absence of these features in partially premixed combustion [28]. Few studies have proposed techniques based on flame emissions [8,29] and adopted nonlinear dynamics [9] and data-driven approaches [10,30–32] to detect LBO for hydrocarbon fuel combustion of different degrees of fuel-air mixing.

With the recent advancements of machine learning approaches, researchers proposed a computationally efficient strategy based on back propagation neural networks [33] to predict fuel ignition delay. The back propagation neural network approach was optimized by genetic and mind evolution algorithm to understand thermophysical properties of hydrocarbon fuels [34]. Online LBO prediction frameworks should simultaneously be accurate and computationally fast. While some studies have concentrated on computational time for LBO prediction [9,13,32], a recent study showed that an approach based on Support Vector Machine may predict the beginning of LBO a few milliseconds before it actually occurs [35]. Based on existing studies of LBO detection, further exploration is needed to develop a technique that can detect LBO earlier allowing the controller a sufficient lead time to adjust fuel-air ratio. It is important to simultaneously consider the aspects of prediction accuracy, computation time, and robustness. While the application of deep learning models to detect combustion instability has started recently [36–42], there has been no work till now on early LBO prediction using deep learning. In the present work, we propose a deep learning approach based on Long Short Term Memory (LSTM) to predict an imminent LBO in a robust manner. We formulate the problem in such a way that LBO prediction performance can be tested for robustness. We compare the performance of the proposed framework with three baseline methods in terms of prediction accuracy and computation time. The results demonstrate that the proposed framework is accurate, computationally tractable and robust to generalize well for test datasets from different protocols. We consider the aspects of prediction accuracy, computation time, and robustness in developing the proposed approach.

We summarize the contributions of this work as follows:

- To the best of our knowledge, this is the first work using deep learning to predict lean blowout in combustion systems.

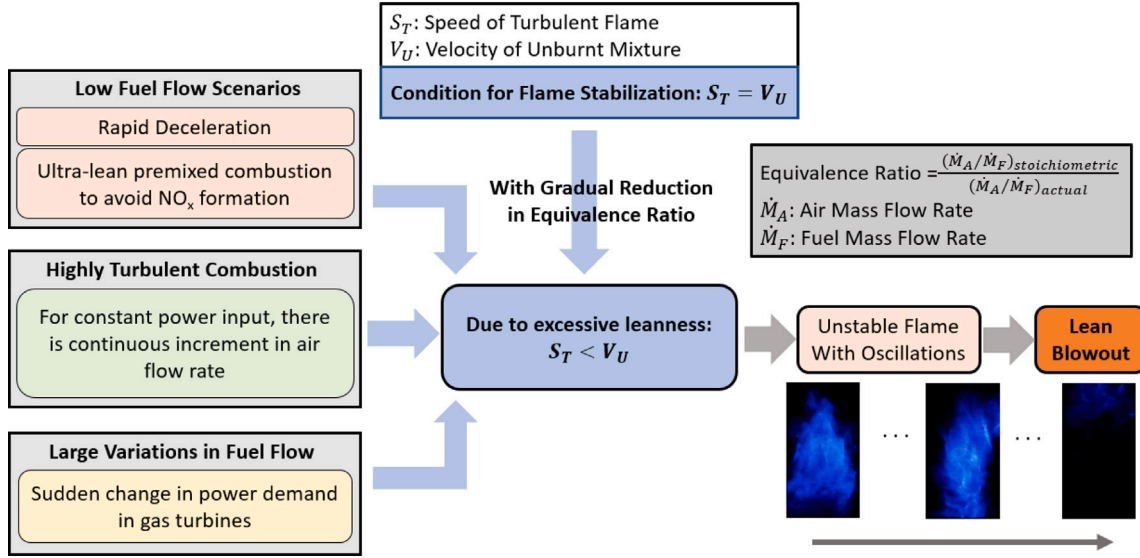


Fig. 1. Illustration of the concept of lean blowout (LBO) in combustion systems.

- Our proposed LBO prediction framework is dependent on pressure time-series data, which is a much convenient way of collecting data from real world combustors. Our training scheme is facilitated by the use of domain knowledge based labeling technique, which is based on flame images.
- To show the robustness of our model, we test in different conditions near the lean blowout regime and use three different baseline models for comparison in terms of accuracy and computation time.
- Our proposed deep learning approach is accurate, computationally faster and captures the transition to LBO regime better than the baseline methods. It shows the potential to be implemented as an online predictor to prevent the adverse effects of LBO in combustion systems.

2. Details of the model combustor

The model combustor used in this work is the same as that used in some previous studies [8,9,24]. An illustration of the experimental setup is provided in Fig. 2. The experimental setup primarily consists of three major components: a premixing chamber, a combustion chamber, and an exhaust chamber or outlet. The premixing chamber is constructed to control the quality of oxidizer-fuel mixing [24]. The chemical reaction takes place in the combustion chamber.

The premixing chamber, shown in Fig. 2(a), has the bottommost entry point for the oxidizer (air is used), and the other entry points from upstream to downstream of the premixer are used for the fuel. Therefore, the premixing chamber allows the mixing of fuel and oxidizer at different mixing lengths L_F [24]. At each entry point, there is a provision of four inlets (90° apart from each other) to reduce the asymmetry in the flow. After the topmost entry point, a swirler (swirl number = 1.26, the calculation has been shown in an earlier study [24]) is provided to improve the mixing, thereby enhancing the stability of combustion [43]. The vane angle of the swirler is 60° to the axial direction. After the swirler, the sudden expansion in the flow area of the dump plane (Fig. 2(b)) slows down the flow velocity and generates re-circulation regions, ensuring continuous feedback of heat radicals to the incoming mixture. To provide optical accessibility, the combustion chamber (Fig. 2(c)) is surrounded by a quartz tube of length 200 mm and an outer diameter of 65 mm. The burnt gases come out of the combustion chamber through the exhaust chamber.

3. Methods

In this section, first, we elaborate on the details of dataset collection (Section 3.1) followed by label annotation technique (Section 3.2), which is based on domain knowledge. We dive deeper into the problem formulation in Section 3.3 and thereafter, in Section 3.4, we describe the architecture of our proposed deep learning approach. In Section 3.5, we discuss about the details of baseline methods used for performance evaluation.

3.1. Dataset collection

We collect data in multiple protocols and for each protocol, we approach the lean blowout condition by decreasing the fuel flow rate with a constant air flow rate. In other words, we move towards LBO with a variation of equivalence ratio (Φ) from the stoichiometric condition (where, $\Phi = 1$). We measure the fuel and air flow rate using mass flow controllers (Alicat Scientific, MCR series, measuring range of 0–250 SLPM for fuel and 0–1500 SLPM for air). For each equivalence ratio, an audio sensor (sampling frequency of 48 kHz) is used to record the acoustic time series, which captures the pressure fluctuations in the system. We also capture flame images to understand the combustion behavior while annotating the acoustic time series for each experiment.

As mentioned earlier, we keep the air flow rate constant at 70 SLPM, 75 SLPM, 80 SLPM, 85 SLPM, and 90 SLPM respectively for five different protocols in our data collection procedure. In the experimental setup (Fig. 2), we use the third fuel entry point (towards the downstream of the air entry point) as the fuel port, ensuring partial premixing of the fuel and oxidizer. For each protocol, we record acoustic pressure time series at different fuel flow rates. For example, in the 70 SLPM protocol, with the air flow rate fixed at 70 SLPM, we generate a set of quasi-static time series datasets corresponding to different fuel flow rates. The equivalence ratio (Φ) where lean blowout occurs is referred to as Φ_{LBO} . We represent each quasi-static dataset by Φ/Φ_{LBO} ratio where lean blowout implies $\Phi/\Phi_{LBO} = 1$ and higher values of Φ/Φ_{LBO} indicate regimes away from lean blowout.

3.2. Labels annotated using domain knowledge

The stability of a combustion process generally depends on factors like hydrodynamics of the flow [44], acoustic pressure fluctuations [45]

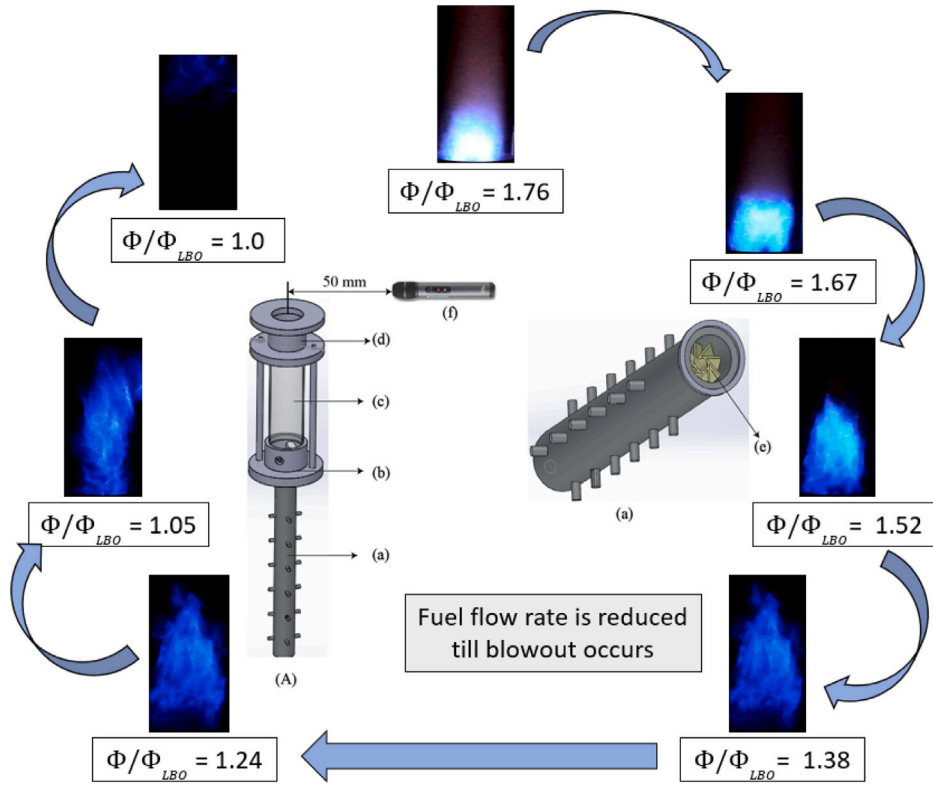


Fig. 2. The experimental setup is shown along with example images for different values of Φ/Φ_{LBO} . The fuel flow rate is reduced till blowout occurs at $\Phi/\Phi_{LBO} = 1$. The accessories of the experimental setup (A) are: (a) premixing chamber, (b) dump plane, (c) combustion chamber covered with quartz tube, (d) exhaust chamber and (e) swirler having 7 blades. Microphone (f), kept at a distance of 5 cm from the centre of the combustor rig, is used for capturing the timeseries data (combustion sound).

and heat release rate oscillations (influenced by chemical kinetics or combustion reaction rates) [24]. A discrepancy in these factors may lead to unusual behavior in the combustion zone. Based on domain knowledge and by utilizing sample flame images, we can understand the transition in combustion regimes. From flame images, we can annotate each Φ/Φ_{LBO} quasi-static dataset as healthy or unhealthy. Fig. 2 demonstrates sample flame images at different Φ/Φ_{LBO} . Annotation ‘healthy’ refers to conditions far away from the LBO regime with Φ/Φ_{LBO} values higher than the transition Φ/Φ_{LBO} . Conditions with Φ/Φ_{LBO} lower than the transition Φ/Φ_{LBO} are annotated ‘unhealthy’ and are therefore near the LBO regime.

From the sample set of flame images shown in Fig. 2, we observe that near the stoichiometric condition ($\Phi/\Phi_{LBO} = 1.76$), the flame adheres to the dump plane (Section (b) of experimental setup (A)) and therefore the combustion process looks stable. This is usually observed when there is a strong equilibrium between the flame speed and the upcoming unburnt mixture [4,24]. The flame manages to survive at the inlet of the combustion chamber in the presence of high incoming flow ($Re \sim 3625$). This flame attachment exists till $\Phi/\Phi_{LBO} = 1.52$ and we observe a significant variation in combustion dynamics at $\Phi/\Phi_{LBO} = 1.38$, when the flame remains at a standoff distance from the chamber inlet. With the air flow rate fixed, as the fuel flow rate is decreased to lower values, the chemical reaction rate also decelerates which in turn reduces the heat release rate of combustion [4,21] and impacts the attachment between flame base and combustion inlet.

The bifurcation in the flame behavior at $\Phi/\Phi_{LBO} = 1.38$ can be attributed as transition [9,24]. Therefore, the flame behavior for $\Phi/\Phi_{LBO} > 1.38$ can be referred to as ‘healthy’. With further reduction in Φ/Φ_{LBO} , the flame dynamics gets sufficiently weak and irregularities in behavior are noticed in $\Phi/\Phi_{LBO} = 1.24$ and $\Phi/\Phi_{LBO} = 1.05$ as demonstrated in Fig. 2. As the condition approaches lean blowout, the rapid fluctuations start in the flame base with local extinction and re-ignition events [8,24] ultimately leading to LBO, represented as

$\Phi/\Phi_{LBO} = 1$. The flame behaviors at $1 \leq \Phi/\Phi_{LBO} < 1.38$ are referred to as ‘unhealthy’ states.

It is important to note here that the transition Φ/Φ_{LBO} value of 1.38 is specific to the data collection protocol shown for demonstrative purpose in Fig. 2. This transition Φ/Φ_{LBO} value is not protocol-agnostic and can only be determined through an offline analysis of quasi-static flame image datasets of each protocol as explained above. The process of annotating healthy and unhealthy states is entirely manual and distinct across protocols. We require annotations from one protocol to build the LBO prediction pipeline and the annotations of other four protocols are used only for the purpose of performance evaluation. We discuss more about this later in the paper. The transition from healthy to unhealthy state can occur rapidly in real combustors and the probability of avoiding LBO becomes low once the flame enters into the critical regime (very close to LBO) [9]. Therefore, it is preferable to predict the occurrence of LBO early to allow a sufficient lead time in adopting precautionary measures.

3.3. Training, reference and test datasets

Out of five protocols, keeping one protocol (air flow rate = 90 SLPM) as the reference protocol, we use the other four protocols as test protocols. For the reference protocol, we collect quasi-static data at following Φ/Φ_{LBO} conditions:

- 90 SLPM: $\Phi/\Phi_{LBO} = [1, 1.142, 1.214, 1.285, 1.357, 1.428, 1.500, 1.571, 1.642, 1.714, 1.785]$

As mentioned in Section 3.2, the transition Φ/Φ_{LBO} will vary with protocols. Determining the transition Φ/Φ_{LBO} for a protocol requires offline analysis of flame images after LBO has already occurred (i.e. after we already know the equivalence ratio Φ_{LBO}). Therefore, such an approach is not suitable for an online LBO prediction. But, we can utilize the annotated transition state information for a reference

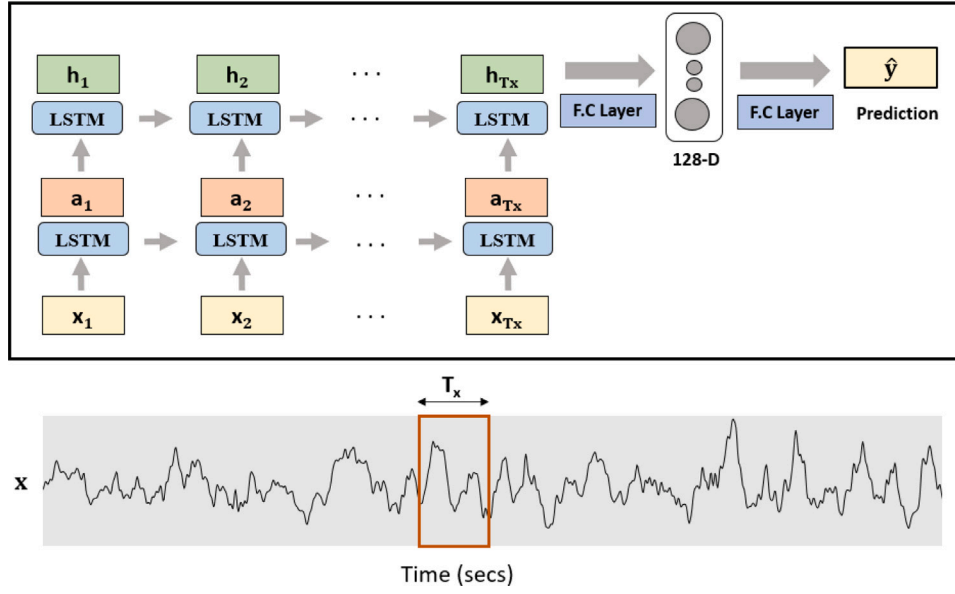


Fig. 3. Illustration of the proposed LSTM based deep learning model. Given the information for T_x input time steps, the model learns to predict the immediate future time step $T_x + 1$.

protocol and train a deep learning model to compute a transition state Φ/Φ_{LBO} metric. Once we have a trained model and the transition state metric, it should be sufficient for online LBO prediction. Therefore, in test scenarios, by utilizing the trained model and comparing against the transition state metric, we can directly predict a state as ‘healthy’ or ‘unhealthy’.

To test the performance of our proposed approach, we use quasi-static datasets from four test protocols:

- 70 SLPM: $\Phi/\Phi_{LBO} = [1, 1.076, 1.153, 1.23, 1.307, 1.384, 1.461, 1.538, 1.615, 1.692, 1.769]$
- 75 SLPM: $\Phi/\Phi_{LBO} = [1, 1.071, 1.143, 1.214, 1.285, 1.357, 1.428, 1.5, 1.571, 1.643, 1.714, 1.785]$
- 80 SLPM: $\Phi/\Phi_{LBO} = [1, 1.066, 1.133, 1.2, 1.266, 1.33, 1.4, 1.466, 1.533, 1.6, 1.67]$
- 85 SLPM: $\Phi/\Phi_{LBO} = [1, 1.071, 1.143, 1.214, 1.285, 1.357, 1.428, 1.5, 1.571, 1.643, 1.714, 1.785]$

3.4. LSTM based deep learning model

We denote the univariate time series input as $\mathbf{X} = [x_1, x_2, \dots, x_T]^T \in \mathbb{R}^T$, where T denotes the total length of the input time series. We divide the entire dataset into consecutive time windows — each window of length T_x . The consecutive windows can be defined as: $[x_1, \dots, x_{T_x}]$, $[x_{T_x+1}, \dots, x_{2T_x}]$, $\dots$, $[x_{(T-T_x+1)}, \dots, x_T]$. We denote the k th time window as $\mathbf{x}^k = [x_k, \dots, x_{(k+T_x-1)}]^T \in \mathbb{R}^{T_x}$, $k \in \{1, 2, \dots, (T - T_x + 1)\}$. With a slight change of notation, the time series of the k th time window can be denoted by $\mathbf{x}^k = [x_1^k, x_2^k, \dots, x_{T_x}^k]^T \in \mathbb{R}^{T_x}$. We follow this pre-processing technique for all the quasi-static time series datasets from reference and test protocols.

Recurrent Neural Network (RNN) based models have proved to be effective in different applications including machine translation [46], speech recognition [47], performance monitoring of engineering systems [39], financial markets [48] and healthcare [49]. Recurrent Neural Networks (RNNs) are efficient in learning the temporal dependencies from time series data for accurate forecasting [50]. RNNs are trained using the error backpropagation algorithm in which the error gradients are propagated through the unrolled temporal layers. In an RNN block, at each time-step, the input is combined with the hidden state vector using a learned function to produce a new state vector.

The architecture of a RNN allows the information from the previous time-steps to persist by passing the hidden state from one step of the network to the other. To overcome the problem of vanishing gradients of RNNs for long sequences [51,52], an effective RNN architecture is Long Short Term Memory (LSTM) [53]. LSTMs, more effective in capturing long range dependencies from time series data, also mitigate the vanishing gradient problems leading to stable model training. In an LSTM block, the input, output, and forget gates regulate the addition of any information to the cell state, which is designed to capture information across longer sequences.

In this work, we develop an LSTM based deep learning model, illustrated in Fig. 3. With a time series of T_x time steps as input, the model learns to predict the immediate future time step $T_x + 1$. The first LSTM layer of the model reads the input sequence in order from x_1 to x_{T_x} to compute the hidden states $[a_1, a_2, \dots, a_{T_x}]$. Through experiments, we find the optimal value of T_x to be 50.

In the first LSTM layer, at time-step t , apart from the input x_t , an LSTM block takes in the cell state c_{t-1} and hidden state a_{t-1} of the previous time-step. Forget gate layer (σ_1), the first step of an LSTM block, decides information to be deleted from the cell state c_t :

$$e_t = \sigma_1(W_e^T[a_{t-1}; x_t] + b_e)$$

The input gate controls the amount of new information to be added to the cell state. The input gate layer (σ_2) is followed by a tanh layer to generate new values for the cell state:

$$r_t = \sigma_2(W_r^T[a_{t-1}; x_t] + b_r)$$

$$\tilde{c}_t = \tanh(W_c^T[a_{t-1}; x_t] + b_c)$$

Next, the new cell state is computed:

$$a_t = e_t \times c_{t-1} + r_t \times \tilde{c}_t$$

The output gate (σ_3) retrieves relevant information to produce output from an LSTM block and new hidden state is computed:

$$o_t = \sigma_3(W_o^T[a_{t-1}; x_t] + b_o)$$

$$a_t = o_t \times \tanh(c_t)$$

The updated hidden states a_t and c_t act as input for the next LSTM block.

The sequence of hidden states from the first LSTM layer $[a_1, a_2, \dots, a_{T_x}]$ acts as the input to the second LSTM layer which

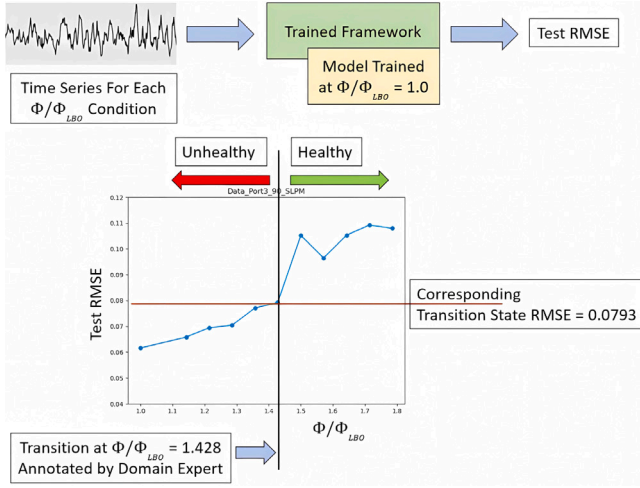


Fig. 4. Quasi-static time series datasets from reference protocol (90 SLPM) used to compute transition state RMSE (0.0793) corresponding to the transition $\Phi/\Phi_{LBO} = 1.428$. We use the LSTM model trained at $\Phi/\Phi_{LBO} = 1$.

generates the hidden states $[\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_{T_x}]$. Like the first layer, similar operations happen inside an LSTM block of the second layer with $\mathbf{h}_{t-1}$, the hidden state from time-step $t-1$, and $\mathbf{a}_t$ as inputs at time-step t . With an optimal hidden state dimension of 128 determined through experiments, the number of trainable parameters in the first and second LSTM layer are around 66k and 131k respectively. The state $\mathbf{h}_{T_x}$, capturing the summarized temporal information of the input time series, is passed through two fully connected layers to predict $\hat{y}$, the next time step. The fully connected layers have approximately 17k trainable parameters, contributing to a total of around 214k trainable parameters for the entire model.

We build our LBO prediction pipeline utilizing time series data of different conditions in the reference protocol (90 SLPM), described in the previous sections. For the reference protocol, through an offline analysis based on domain knowledge, we annotate each Φ/Φ_{LBO} quasi-static dataset including the transition and $\Phi/\Phi_{LBO} = 1$ condition, where LBO occurs. In our proposed deep learning approach, we need to train the LSTM model using time series from only one condition, $\Phi/\Phi_{LBO} = 1$ of the reference protocol. Therefore, after training, the trained model only has knowledge about the temporal dynamics of the $\Phi/\Phi_{LBO} = 1$ condition.

Keeping aside 10% of the data, the training and validation sets together comprise the rest 90% of the $\Phi/\Phi_{LBO} = 1$ time series sequence. From the train-validation set, 70% of data is used for training the LSTM model. The training dataset is used to fit scalars to transform features to a range of (0, 1). The scalars are saved to be utilized later for inference to scale each feature before input to the trained model. We divide a time series into consecutive time windows — each window represented by T_x time-steps for input and the value of the immediate future time-step for desired output (to compute RMSE). We perform experiments to select an optimal set of hyper-parameters based on the validation set performance. We train the model on an NVIDIA Titan RTX GPU for 50 epochs with a batch size of 512, learning rate of 0.001 using the Adam optimizer [54].

3.5. Baseline methods

Recurrent Neural Network (RNN): As a deep learning based baseline model, we use a Recurrent Neural Network (RNN) modeling approach developed by substituting LSTM units in Fig. 3 with RNN units. We train the RNN model in similar settings as with the LSTM model.

The illustration of the RNN baseline model is provided in the supplementary material. The model consists of two stacked RNN layers.

For the first layer, at time-step t , the input x_t and previous hidden state $\mathbf{a}_{t-1}$ is passed to an RNN block, which computes $\mathbf{a}_t$ using a non-linear mapping as follows:

$$\mathbf{a}_t = \tanh(W_a^T [\mathbf{a}_{t-1}; x_t] + b_a)$$

This updated hidden state $\mathbf{a}_t$ is utilized in the next time-step. After reading the input sequence in order from x_1 to x_{T_x} , the first RNN layer returns a sequence of hidden states $[\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_{T_x}]^T$, which acts as input to the second RNN layer. For the second RNN layer, at time-step t , the inputs to the RNN block are $\mathbf{h}_{t-1}$, the hidden state at time-step $t-1$, and $\mathbf{a}_t$. The updated hidden state $\mathbf{h}_t$ is computed as:

$$\mathbf{h}_t = \tanh(W_h^T [\mathbf{h}_{t-1}; \mathbf{a}_t] + b_h)$$

The hidden state $\mathbf{h}_{T_x}$ of the final time-step is passed to the fully connected layers. The number of trainable parameters in the first and second RNN layer are around 16.6k and 32.8k respectively. The total number of trainable parameters in the RNN model is approximately 66k.

Hidden Markov Model (HMM): We use Hidden Markov Model (HMM) as a machine learning based baseline modeling approach. Recently, Hidden Markov Models (HMMs) have been utilized for early prediction of a lean blowout from chemiluminescence time series data [13]. HMMs, probabilistic frameworks for recognizing patterns in stochastic processes, have proved to be successful in speech recognition problems [55]. In an HMM, a sequence of observations is generated from a sequence of internal states. The likelihood of the observations depends on the states which are hidden. The transitions between the hidden states are assumed to follow a first-order Markov chain. An HMM is defined by the start probability vector Π , transition probability matrix A , and emission probability of an observation Θ_i .

In the HMM training process, we utilize the $\Phi/\Phi_{LBO} = 1$ dataset from the reference protocol using the same training set (90%) used for training the LSTM model. The number of states with the least Bayesian Information Criterion is chosen as the optimal number. After getting the optimal HMM, we divide the training dataset into consecutive time windows of length T_x to compute the log-likelihood for each window in the training set. During inference, for a time window k , we first predict the log-likelihood using the trained HMM. The log-likelihood value is then compared with the log-likelihood values of the training set windows to find the closest window j . We can consider the training set window $\mathbf{x}^j = [x_1^j, x_2^j, \dots, x_{T_x}^j]^T$ as most similar to the inference window $\mathbf{x}^k = [x_1^k, x_2^k, \dots, x_{T_x}^k]^T$. Next, we compute the change as $x_{T_x+1}^j - x_{T_x}^j$ and add this change to $x_{T_x}^k$ to get prediction $\hat{y}_k$ for the k th window.

Translational Error: As another baseline method, we use translational error [56], an existing and well-known method in this domain. Translational error has been previously proposed as a suitable candidate for predicting lean blowout [56]. It is a nonlinear dynamic tool capable of identifying the dissimilarity in the directions of neighboring trajectories. Mathematically, it is expressed as:

$$E_{trans} = \frac{1}{K+1} \sum_{k=0}^K \frac{\|v(t_k) - \bar{v}\|^2}{\|\bar{v}\|^2}, \quad (1)$$

where, $\bar{v} = \frac{1}{K+1} \sum_{k=0}^K v(t_k)$. Here, $v(t_k) = P'(t_k + \tau_d) - P'(t_k)$. In Eq. (1), K indicates the number of neighboring points of $P'(t_i)$. We use $K = 5$, as suggested in a previous study [56].

The median of E_{trans} values is estimated for 100 randomly chosen phase space vectors, $P'(t_i)$. Therefore, the computation of translational error has a randomness factor, for which we use three independent computations in the case of the test dataset to report the mean and standard deviation. And to compare the computation time with other methods, we use the mean computation time from the three computations. Two other important parameters are used in constructing the phase space vectors: time delay and optimum embedding dimension. These are derived after experiments with the training dataset $\Phi/\Phi_{LBO} = 1$ from the reference protocol.

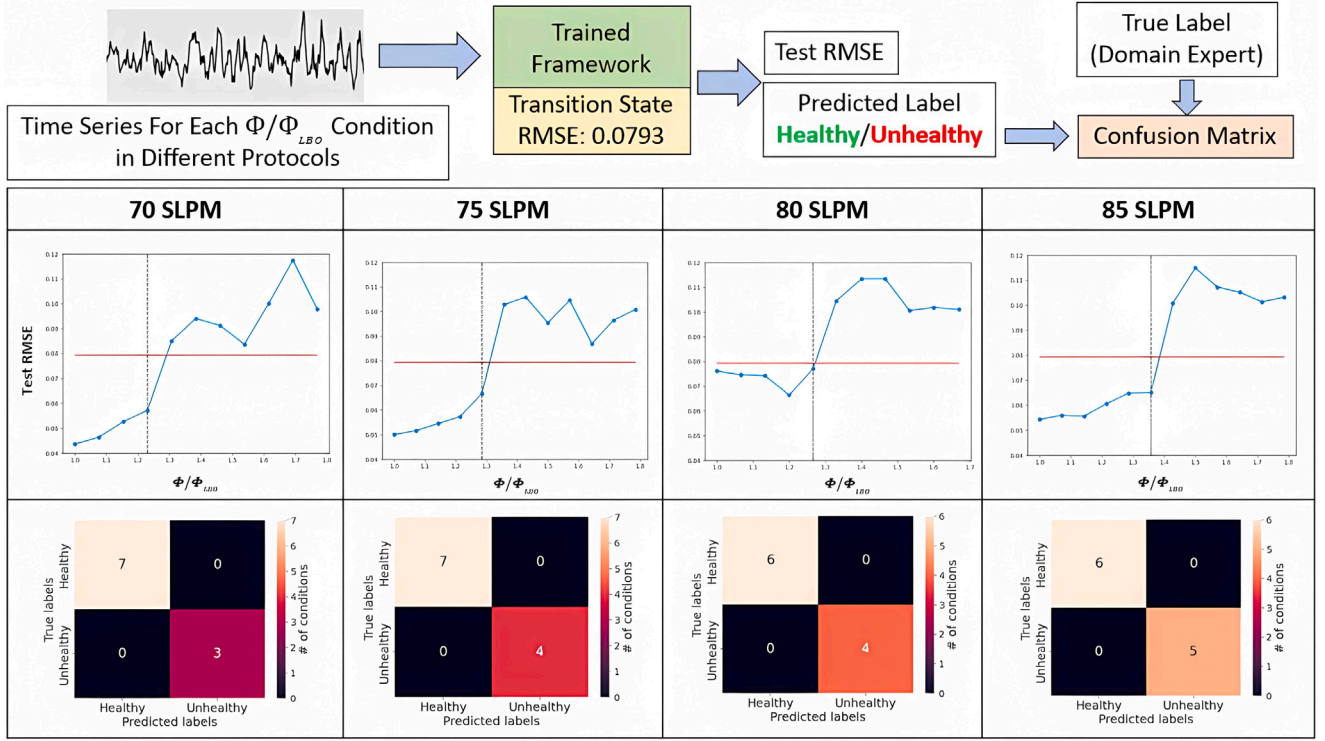


Fig. 5. Performance on quasi-static time series datasets from four test protocols using the trained LSTM model and corresponding transition state RMSE (0.0793). For each protocol, the test RMSE values are plotted for different Φ/Φ_{LBO} conditions and confusion matrix is computed. The transition state RMSE (0.0793) is shown with a red horizontal line, and vertical black dotted lines represent the corresponding ground truth transition states, annotated by domain experts. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Results and discussion

4.1. Transition state metric from the reference protocol

We utilize the quasi-static datasets from the reference protocol (90 SLPM) to compute a metric for transition. There are eleven quasi-static datasets for the 90 SLPM protocol with $\Phi/\Phi_{LBO} = [1, 1.142, 1.214, 1.285, 1.357, 1.428, 1.500, 1.571, 1.642, 1.714, 1.785]$. The transition occurs at $\Phi/\Phi_{LBO} = 1.428$, determined through our annotation process explained in Section 3.2. We train the LSTM model using 90% of time series data from the $\Phi/\Phi_{LBO} = 1$ condition. Using the unseen sequence (10%), we perform inference with the trained model, computing root mean square error (RMSE) between predicted and target values. As mentioned earlier, in the pre-processing step, we divide a time series sequence into consecutive time windows, each with length T_x . For example, if there are K number of time windows in the test sequence, RMSE is computed as:

$$RMSE = \sqrt{\frac{1}{K} \sum_{k=1}^K (\hat{y}_k - y_k)^2}$$

We present the approach of determining transition state RMSE in Fig. 4. As the model has been trained on the $\Phi/\Phi_{LBO} = 1$ condition, we observe the least test RMSE for this condition. With the same pre-processing approach and using the trained $\Phi/\Phi_{LBO} = 1$ model, we perform inference for all the other Φ/Φ_{LBO} conditions. We compute a test RMSE value for each Φ/Φ_{LBO} - the plot is shown in Fig. 4. The error increases as we move away from the LBO regime and keeps increasing beyond the transition state. Unlike healthy conditions with higher error values, unhealthy conditions are characterized by lower RMSE scores implying closer proximity to LBO. In an online testing scenario, a threshold value of RMSE is needed to classify a condition as healthy or unhealthy.

For the annotated transition state ($\Phi/\Phi_{LBO} = 1.428$) of the reference protocol, the computed test RMSE value is 0.0793. We can consider this transition state RMSE value as a threshold to classify a condition as healthy or unhealthy, as demonstrated in Fig. 4. If the RMSE is more than 0.0793 for a test condition, we can predict the condition as healthy. Otherwise, if the RMSE error is less than 0.0793, the condition is considered unhealthy and will require preventive measures to prevent a lean blowout. The reference protocol results for the RNN model, HMM model, and translational error method are provided in supplementary material. As the translational error method has a randomness factor, we perform three independent computations and compute the mean and standard deviation. With the LSTM model trained and the transition state metric obtained, next, we test the performance of this LBO prediction approach on four test protocols, each consisting of a set of quasi-static time series datasets.

4.2. Performance on test protocols

We test the performance of the proposed approach on quasi-static datasets from four protocols (70 SLPM, 75 SLPM, 80 SLPM, 85 SLPM). For preprocessing, as described before, we divide a time series into consecutive time windows — each window represented by $T_x = 50$ time-steps (to be used as model input) and the value of the immediate future time-step (to be used for comparing against model prediction to compute RMSE). For each Φ/Φ_{LBO} condition (quasi-static dataset), the trained LSTM model takes in time series data as input after required pre-processing, and an RMSE value is computed using the predictions from the model and the actual values. As the trained LSTM model has the knowledge of only the LBO condition, generally the RMSE value will be higher if a condition is far away from LBO and lower if that condition is near the LBO regime. Therefore, a test protocol comprising a set of conditions is represented by multiple RMSE values. When the



Fig. 6. Overall confusion matrix for each method considering all the test protocols.

RMSE values are plotted for varying conditions (Φ/Φ_{LBO}) in a protocol, we observe a common trend for all four protocols — the RMSE value decreases as we move closer to LBO ($\Phi/\Phi_{LBO} = 1$), demonstrated in Fig. 5. Initially, for higher Φ/Φ_{LBO} values, with elevated levels of RMSE, a protocol is in a healthy state. As the Φ/Φ_{LBO} changes, at a certain condition, the RMSE drops below the pre-determined transition state RMSE (0.0793), indicating entry to the unhealthy regime — closer to LBO. In Fig. 5, we denote the transition state RMSE with a horizontal red line. Accurately predicting this change is extremely important for the real-world use of these prediction systems. A condition is predicted as healthy or unhealthy depending on whether the RMSE value is higher or lower than the transition state RMSE value.

We can compare the predicted labels against the actual labels as we know the ground truth annotations of transition states, shown as vertical black dotted lines in Fig. 5. When we compute a confusion matrix to evaluate performance for each protocol, we find that the proposed LSTM-based framework achieves 100% accuracy, correctly predicting healthy and unhealthy states for all four test protocols. We provide test protocol results for the RNN model, HMM model, and translational error method in the supplementary material.

4.3. Performance comparison in terms of prediction accuracy

To summarize the comparative performance across all the test protocols, we demonstrate the overall confusion matrix for each method in Fig. 6. HMM correctly predicts unhealthy conditions but falsely identifies most healthy conditions as unhealthy, leading to higher false positives. Here, false positive refers to falsely predicting a negative (or healthy) condition as positive (or unhealthy). Unlike HMM, when we use the translational error method as the LBO prediction tool, we observe that it falsely predicts half of unhealthy conditions as healthy while showing 100% accuracy for healthy conditions. Therefore, the translation error method leads to false negatives — falsely identifying unhealthy (or positive) conditions as healthy (or negative). High false negatives can exemplify the difficulty of online LBO prediction, as it can adversely affect decision-making to prevent such scenarios and ensure smooth engine operation. The randomness factor in computing translational error can also be a concern in its use as an LBO prediction tool. The major disadvantage of HMM lies in false positives that can lead to false alarms, which is highly undesirable in real-time control.

While HMM suffers from false positives and the translational error method leads to false negatives, LSTM and RNN demonstrate 100% prediction accuracy, according to confusion matrices in Fig. 6. The deep

learning methods outperform HMM and translational error method, making our proposed approach adept in capturing transitions in combustion systems and timely identifying any imminent dangers from LBO.

4.4. Performance comparison in terms of computational efficiency

Not only accuracy but also computational efficiency is a crucial factor in the performance evaluation of methods for online LBO prediction. We compare the error computation time of each method using the same computation resource (CPU) for Φ/Φ_{LBO} conditions in the 70 SLPM test protocol. From Fig. 7, we observe that the computation times for HMM are much higher than those for translational error, RNN, and LSTM. The computation times for the deep learning models LSTM and RNN are quite similar, with the LSTM model demonstrating slightly lower computation times. Similar trends are seen for the other three test protocols. The deep learning models, outperforming other methods in terms of both prediction accuracy and computational efficiency, show the potential for implementation in real-time detection and control of lean blowout. In this work, while the accuracy of LSTM and RNN is same, the LSTM model shows marginally superior inference speed compared to the RNN model. The performance of deep learning models can slightly vary due to numerical reasons. It is crucial to conduct benchmarking on specific datasets for both the LSTM and RNN model to accurately assess inference speeds before deployment. As LSTMs tend to offer better performance on tasks that involve learning long-term dependencies, we recommend LSTM as the preferred option for online LBO prediction.

5. Conclusion

The adaption of lean combustion technology helps to reduce the NO_x emissions, although it often leads to the occurrence of lean blowout due to the extreme leanness of the fuel-air mixture. Such an extinction of flame can lead to sudden loss of power in combustion systems, causing unwanted adverse events. Therefore, developing effective and feasible frameworks for real-time accurate detection and control of LBO is essential. With this motivation, in this paper, we develop a deep learning-based framework for predicting lean blowout in combustion systems.

In the present work, we utilize a trained LSTM model and a transition state metric to identify a combustion state as healthy or unhealthy. The proposed framework based on LSTM is robust enough to detect the transition to the LBO regime in unseen test protocols. Comparing performance against baseline methods, we find that both deep learning approaches outperform HMM and translational error method in terms of accuracy and computational efficiency. Between the two deep learning approaches, we observe that the LSTM model shows same accuracy but slightly lower computation time than the RNN model. The deep learning models have the same architecture — the RNN model is developed by substituting LSTM units with RNN units. It is essential to benchmark deep learning model architectures for any particular task to determine actual inference speeds. In general, we recommend LSTM as the preferred choice to develop this type of LBO prediction framework due to its ability to capture long-range dependencies and stable training process.

As the first deep learning work on LBO prediction, it shows the potential of harnessing the power of deep learning in real-time performance monitoring of engines. This work encourages us to explore attention-based deep learning frameworks [40,57,58] for LBO prediction in future. A similar modeling approach can be adopted to develop control mechanisms of other cyber-physical systems using time series as a sensing modality.

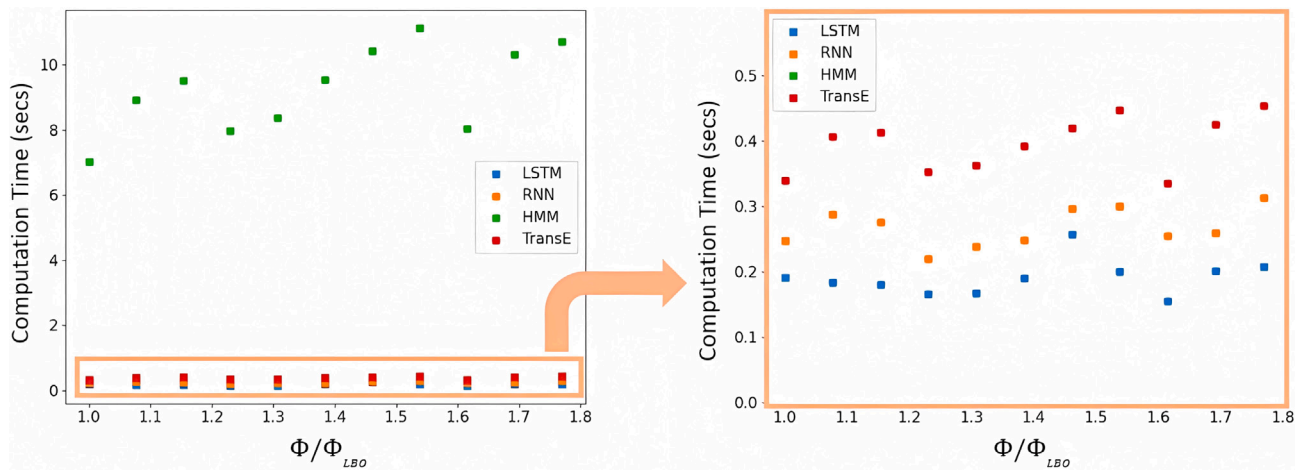


Fig. 7. Comparison of error computation times for LSTM, RNN, HMM, and Translational Error (TransE) using Φ/Φ_{LBO} conditions from the 70 SLPM test protocol. The figure on the right is a zoomed-in view highlighting the computation times from LSTM, RNN, and TransE.

CRediT authorship contribution statement

Tryambak Gangopadhyay: Conceptualization, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing, Formal analysis. **Somnath De:** Conceptualization, Data curation, Validation, Writing – original draft, Writing – review & editing. **Qisai Liu:** Validation, Writing – review & editing. **Achintya Mukhopadhyay:** Project administration, Resources, Supervision, Writing – review & editing. **Swarnendu Sen:** Project administration, Supervision, Writing – review & editing, Resources. **Soumik Sarkar:** Funding acquisition, Investigation, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

See the supplementary material for additional results.

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.egyai.2023.100334>.

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